MIZBAUR RAHMAN ZIHAD

Full-Stack Developer | Next.js | Express.js | TypeScript | PostgreSQL

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CAREER OBJECTIVE

Mathematics student at SUST and Full-Stack Developer specializing in Node.js, Express.js, TypeScript, PostgreSQL (Prisma), and Next.js. Experienced in architecting REST APIs, real-time WebSocket systems, role-based access control (RBAC), and secure payment gateway integrations, paired with responsive Next.js frontends. Combining discrete mathematics logic optimization with modern engineering to build scalable web applications.

TECHNICAL SKILLS

Backend & Systems: Node.js, Express.js (v5), TypeScript, RESTful API Design, Socket.IO (WebSockets), Modular Architecture, Role-Based Access Control (RBAC)

Databases & ORM: PostgreSQL, Prisma 7 ORM, MongoDB, Mongoose, NeonDB, Redis (Caching), Indexing & Aggregation

Auth & Integrations: JWT, Better Auth, Bcrypt, Judge0 CE, SSLCommerz, Cloudinary, Multer, Zod Schema Validation

Frontend: Next.js 16 (App Router), React.js, JavaScript ES6+, Tailwind CSS, HeroUI, DaisyUI, shadcn/ui, Responsive Design

Tools & Cloud: Git, GitHub, Monorepo Architecture, Vite, npm, Postman, Vercel, Render, Netlify

AI & Productivity: LLM Streaming (Gemini, Grok), Google Antigravity, Claude, ChatGPT, AI-assisted debugging & context engineering

PRACTICAL EXPERIENCE

Back-End AI Engineering Intern — *FlyRank AI* | Jul 2026 – Sep 2026 | Certificate Link

- Engineered an AI vision tagging & content matching engine using Node.js (TypeScript), Express, PostgreSQL (`pgvector`), and Gemini AI, generating 768-dimensional vector embeddings for cosine similarity search.
- Implemented a production MismatchGuard safety engine enforcing vision confidence scoring (≥ 0.60) and category contradiction validation to eliminate cross-category matching errors.
- Built robust data scraping and ingestion pipelines featuring Zod schema validation, disk caching, rate limiting, and Vitest test suites.

Complete Web Development Bootcamp — *Programming Hero* | Dec 2025 – Jul 2026

- Earned **Blackbelt Web Developer Distinction** for outstanding engineering performance in Batch 13 across **5,000+** candidates (Cert ID: **L2B7-1126**).
- Engineered **5+ production-ready** full-stack applications mastering Node.js, Express.js, MongoDB, React, Next.js, and secure JWT authentication.
- Implemented protected router states, role-based access control (RBAC), and multi-environment deployment pipelines across Vercel, Render, and Netlify.

Next Level Software Engineering Bootcamp — *Programming Hero* | Apr 2026 – Present

- Mastered advanced TypeScript, modular backend architecture, clean code practices, serverless PostgreSQL via NeonDB & Prisma 7 ORM, Redis caching, Zod validation, Cloudinary image workflows, and SSLCommerz payment gateways.

TECHNICAL PROJECTS

AlgoArena – Competitive Programming Platform (Team Project) | Live App | Client Repo | Server Repo

Tech Stack: Next.js 15, Express 5, TypeScript, MongoDB, Judge0 CE, Socket.IO, JWT, Tailwind CSS

- Architected real-time Socket.IO WebSocket server with JWT authentication handshakes and room broadcasting (`contest:scoreboard`, `leaderboard:update`).
- Integrated sandboxed Judge0 CE supporting Python 3.13, C++ GCC 14, Node 22, and TS 5.6 with execution limits and output normalization algorithms.
- Implemented CP tie-breaking score aggregations (`totalScore` ↓, `problemsSolved` ↓, `penaltySeconds` ↑) and optimized MongoDB `$match` + `$count` pipelines to prevent OOM bottlenecks.

GearUp – Sports & Outdoor Equipment Rental Platform | Live App | Client Repo | Server Repo

Tech Stack: Node.js, Express.js, TypeScript, PostgreSQL, Prisma ORM, SSLCommerz, JWT, Bcrypt, Next.js 16

- Engineered **25+ RESTful API endpoints** for a peer-to-peer equipment rental API supporting a 5-stage order lifecycle state machine (`PENDING` ➔ `APPROVED` ➔ `PAID` ➔ `PICKED_UP` ➔ `RETURNED`).
- Implemented backend booking collision-checking algorithms and multi-role RBAC separating CUSTOMER, PROVIDER, and ADMIN permissions.
- Integrated two-step SSLCommerz payment validation with server-to-server verification APIs and IPN webhooks, securing **100%** of transaction workflows.

ErythroShare – Healthcare & Blood Donation Platform | Live App | Client Repo | Server Repo

Tech Stack: Next.js 16 (App Router), Express.js, MongoDB, Better Auth, Tailwind CSS, HeroUI V2, JWT

- Engineered a multi-role healthcare coordination portal managing permissions for ADMIN, VOLUNTEER, and DONOR users.
- Built protected Express REST API endpoints with session caching and real-time blood request dispatch tracking.
- Implemented donor search indexing by blood group, district, and availability status to streamline emergency requests, cutting dispatch latency by **40%**.

EDUCATION

B.Sc. in Mathematics (3rd Year) – Shahjalal University of Science and Technology (SUST), Sylhet | 2023 – Present

- Core Focus: Analytical Problem Solving, Discrete Mathematics, Logic Optimization, Algorithms.

CERTIFICATIONS & LANGUAGES

Certifications: Back-End AI Engineering Internship Certificate (Credential Link) — FlyRank AI; **Blackbelt Web Developer Distinction & Complete Web Development Certificate** (Credential Link) — Programming Hero (Cert ID: **L2B7-1126**); Next Level Software Engineering Certification (Ongoing).

Languages: Bangla – Native; English – Intermediate Professional Working Proficiency, comfortable with technical documentation, code reviews, and developer communication.